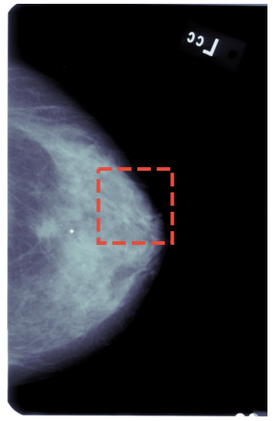
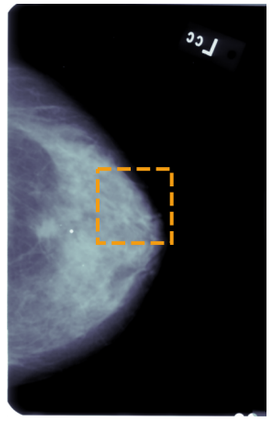
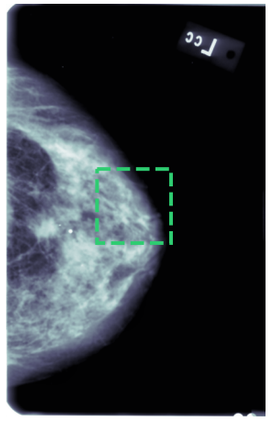
(a) Raw FFDM Input
(Case: P_00001 Malignant)



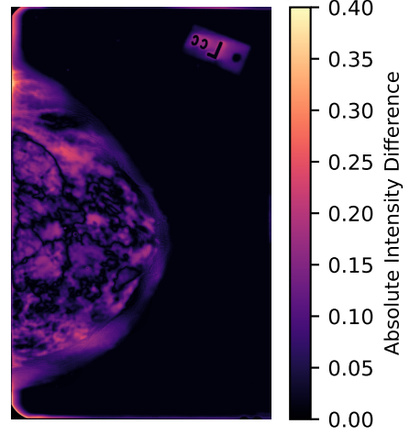
(b) Adaptive Wiener Filter
(5x5 Variance Kernel)



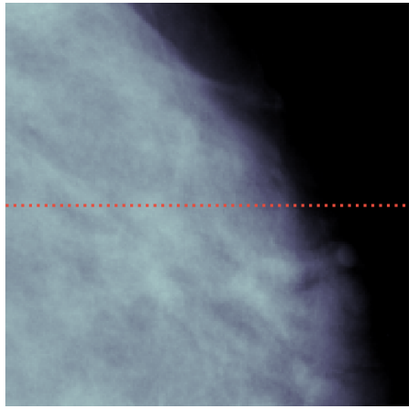
(c) CLAHE Enhanced
(Clip 2.5, 8x8 Grid)



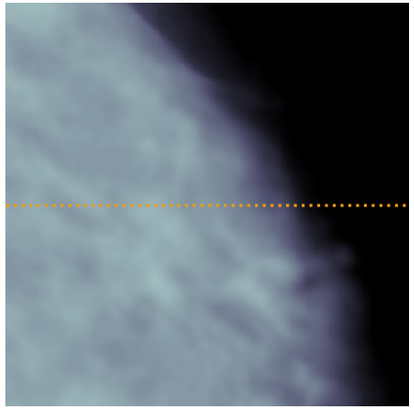
(d) Absolute Residual Map
(|Enhanced - Raw Input|)



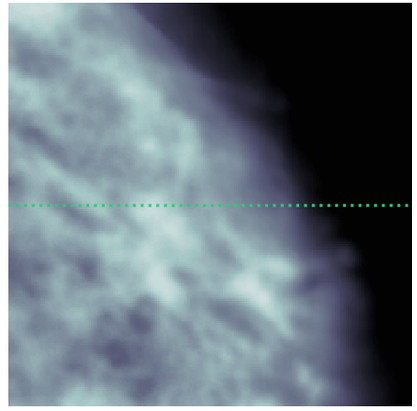
(e1) RoI Core (Raw)



(e2) RoI (Wiener Filtered)



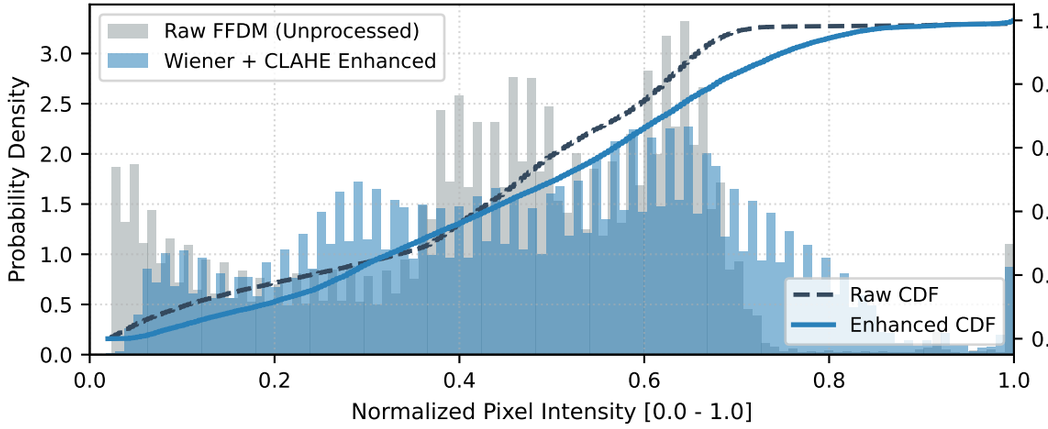
(e3) RoI (Wiener + CLAHE)



(f) Quantitative Preprocessing Metrics

Metric	Raw Image	Preprocess	Delta / Gai
SNR (dB)	7.52 dB	7.86 dB	+0.34 dB
Shannon Entrop	-83.43 bits	-45.48 bits	+37.95 bits
Local Contras	Baseline (1.0)	Enhanced (2.1	42% Contrast
Speckle Noise V	High (Quant	Attenuated (5	8.6% Variance
Boundary Sharp	Blurred/Diffe	Restored Marg	High Gradient

(g) Breast Parenchyma Intensity Distribution and Cumulative Dynamic Range Expansion



(h) Spatial Cross-Section Profile Across Lesion Margin Transect

